Penetration of surface effects on structural relaxation and particle hops in glassy films

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A free surface induces enhanced dynamics in glass formers. We study the dynamical enhancement of glassy films with a distinguishable-particle lattice model of glass free of elastic effects. We demonstrate that the thickness of the surface mobile layer depends on temperature differently under different definitions, although all are based on local structure relaxation rate. The rate can be fitted to a double exponential form with an exponential-of-power-law tail. Our approach and results exclude elasticity as the unique mechanism for the tail. Layer-resolved particle hopping rate, potentially a key measure for activated hopping, is also studied but it exhibits much shallower surface effects.

I. INTRODUCTION

It is well known that a glass former can retain a liquid-like free surface even though its bulk may well have vitrified [1–7]. Of films with free surfaces, a reduction in the glass transition temperature T_g accompanied by accelerated structural relaxation near the surface region has been observed [8–12]. Such surface-enhanced dynamics is confined to a surface mobile layer, over which local structural relaxation is accelerated. Its thickness has been studied in experiments [13–15] and simulations [16–21]. However, estimated values of the thickness and its temperature dependence can vary significantly among different definitions and are not yet fully understood.

A deeper question concerns the mechanism of the enhanced dynamics and the spatial functional form of its penetration into the interior of a film. It was argued that the local structural relaxation time, i.e. α-relaxation time, $\tau_\alpha(z)$ can shed light on the general question of the glass transition [22–29]. Here z is the depth from the free surface of a film. Yet, direct experimental tests of local properties against theoretical predictions are technically difficult, given the limited spatial resolution [23, 30]. Recently, accurate measurements of $\tau_\alpha(z)$ from molecular dynamics (MD) simulations are reported to support a double-exponential form with an exponential-of-power-law tail given by [28]

$$\tau_\alpha^{-1}(z) = (\tau_\alpha^{\text{bulk}})^{-1} \exp\left[a \exp(-z/\xi_0) + b/z\right], \tag{1}$$

more often written as

$$\ln(\tau_\alpha^{\text{bulk}}/\tau_\alpha(z)) = a \exp(-z/\xi_0) + b/z, \tag{2}$$

where ξ_0 measures the mobile layer thickness, $\tau_\alpha^{\text{bulk}}$ is the bulk α-relaxation time, and a and b are constants. Based on an elastically cooperative nonlinear Langevin equation (ECNLE) theory [26], the double-exponential term, found also in early MD simulations [31], can be derived from an empirical layer-by-layer facilitation-like process for surface effects, while the tail can follow from a local softening due to an elastic coupling in three dimensions to the free surface [28]. Although the power-law tail covers only half a decade of scaling and may require further scrutiny, the intriguing two-component form seems evident from the accurate MD results. The functional form of Eq. (2) was claimed as a unique signature for the relevance of elastic field in glass [28]. This echoes a recent increase of interest [32, 33] in the possible dominating role of elasticity in glassy dynamics [34, 35].

Lattice models of glass in general enjoy superior computational efficiency [36, 37]. Here we use a distinguishable particle lattice model (DPLM) [38, 39] to study the relaxation dynamics of supported glassy films. The DPLM has been able to reproduce a variety of characteristic phenomena for bulk samples of glass, including slow relaxation [38], Kovacs paradox [40] and effect [41], kinetic and thermodynamic fragility [39], heat capacity overshoot [42], two-level systems [43], diffusion coefficient power-laws [44], and Kauzmann's paradox [45]. The same model has also successfully simulated surface enhanced mobility [46] and heat capacity [47] of glassy films. The importance of adopting an extensively tested lattice model is to ensure compatibility among explanations of a wide range of glassy phenomena. Mechanisms considered in this study based on the DPLM are thus also consistent with the aforementioned list of characteristics of glass in both bulk and film geometries.

In this paper, we measure and contrast estimates of the thickness of the surface mobile layer from DPLM simulations based on several commonly used definitions. We demonstrate that the local α-relaxation time $\tau_\alpha(z)$ measured for DPLM films can as well be fitted by Eq. (2), despite the absence of elastic effects. Finally, we show that the particle hopping rate from the DPLM admits much weaker surface enhancement than the relaxation time and the implications will be discussed.

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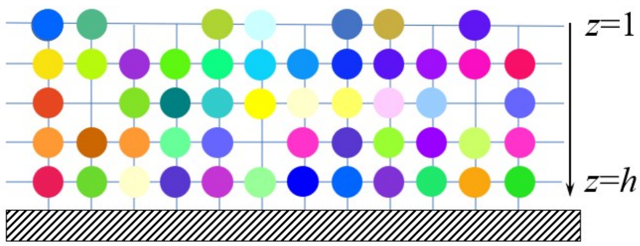

FIG. 1. A schematic drawing of the lattice model of supported film.

II. MODEL

We follow the construction of DPLM films and all model parameters introduced previously for mobility and heat capacity measurements [46, 47]. Specifically, N distinguishable particles of each of their own type populate a square lattice of ribbon geometry, with thickness h and length $L = 1000$, in two dimensions (2D). Each lattice site can be occupied by either a particle or a void (non-occupied), as shown in Fig. 1. We apply periodic boundary conditions along the direction of L and fixed bounding walls along the direction of h. We designate $z = 1$ for the particles in contact with the vacuum represented by the upper wall and $z = h$ for the particles that sit on top of the substrate modeled by the lower wall. The energy of the model is given by

$$E = \sum_{<i,j>} V_{s_i s_j} n_i n_j + \varepsilon_{\rm bot} \sum_{i:z_i=h} n_i + \varepsilon_{\rm top} \sum_{i:z_i=1} n_i \, . \tag{3}$$

Here, the particle type at the site i is labeled s_i. The interaction energy between the particle at the site i and the particle at the adjacent site j is given by $V_{s_i s_j}$. The occupation number $n_i = 1$ at site i if site i is occupied by a particle, otherwise $n_i = 0$ if the site is empty. The z-coordinate (i.e. depth) of site i is given by z_i. We use $\varepsilon_{\rm top} = 1.124$ to account for the excess interfacial energy experienced by particles in $z = 1$ and $\varepsilon_{\rm bot} = -0.5$ for particles in $z = h$, respectively. We sample the interaction energy $V_{s_i s_j}$ from a *a priori* distribution,

$$g(V) = G_0/(V_1 - V_0) + (1 - G_0)\delta(V - V_1), \tag{4}$$

where $V_0 = -0.5$, $V_1 = 0.5$ and $G_0 = 0.7$ are parameters chosen for the study [46].

We simulate void-induced equilibrium dynamics with the Metropolis algorithm. The rate for a particle hopping to an adjacent void is given by

$$w = w_0 \exp\left[-\Delta E \Theta(\Delta E)/k_B T\right], \tag{5}$$

where ΔE is the energy change due to the hop, $\Theta(x) = 1$ if $x > 0$ or $\Theta(x) = 0$ otherwise. We set the attempt frequency $w_0 = 10^6$. We use $k_B = 1$, so the temperature has the same dimension as energy. The adopted algorithm guarantees detailed balance.

We use the swap algorithm (swapping among all particles and voids) to accelerate the equilibration process [44, 48] . More than 10^5 swap attempts per site are performed to equilibrate the energy and depth-dependent particle density. Before taking measurements, another 10^5 swap attempts per site are performed to confirm that equilibrium has been attained. With the parameters chosen above, it has been shown [46] that the particle density at $z > 1$ equilibrates at a constant value of 0.99, and drops sharply to 0.2 at $z = 1$.

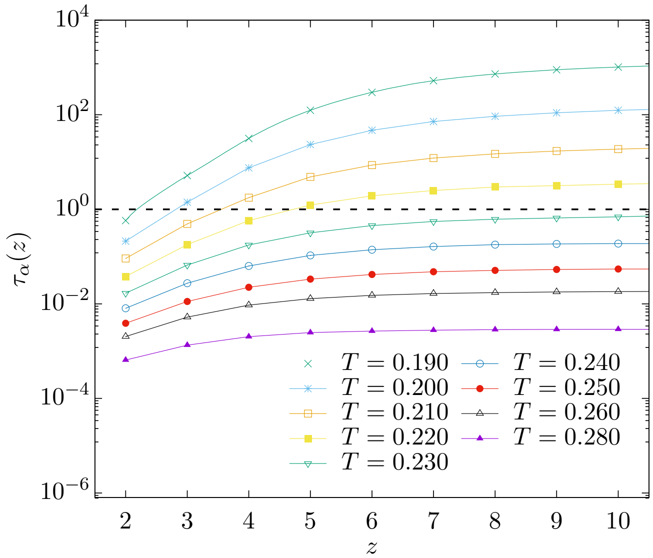

FIG. 2. Structural relaxation time, $\tau_\alpha(z)$, for a film of thickness $h = 45$ at various values of temperature T. The free surface is adjacent to particles at $z = 1$. The dashed line sets an empirical reference observation time $\tau_0 = 1$.

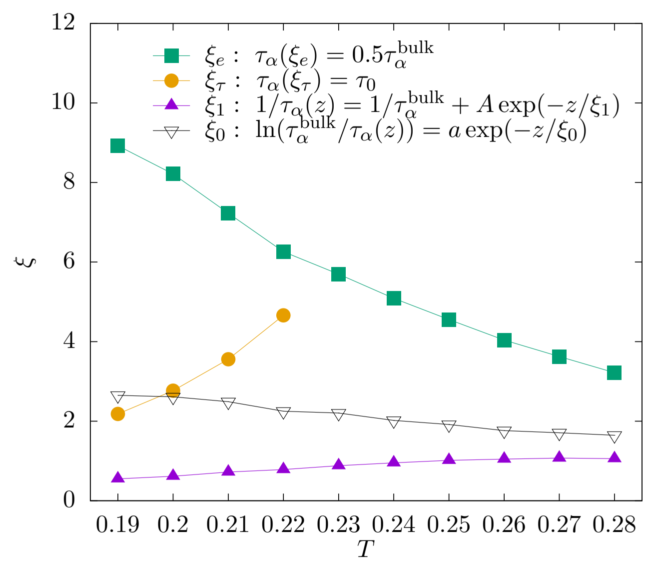

FIG. 3. Thicknesses of surface mobile layer based in different definitions against temperature T for a film of $h = 45$.

III. RELAXATION RATE AND MOBILE LAYER THICKNESS

We measure from DPLM simulations the local overlap function $q(\Delta t, z)$, which gives the probability that a particle in layer z is at the original position after a duration Δt. In Fig. 2, we display the layer-resolved relaxation time $\tau_\alpha(z)$ defined via $q(\Delta t, z) = 1/e$ at $\Delta t = \tau_\alpha(z)$ [46]. As seen, a slope of $\tau_\alpha(z)$ extends towards the bulk and increases as T is reduced from 0.28 to 0.19. This is consistent with MD simulations [16] and signifies the penetration of fast surface dynamics into the interior of the film.

Using $\tau_\alpha(z)$ from Fig. 2, we calculate the mobile layer thickness ξ_0 from the double-exponential term in Eq. (2). Alternatively, we define a different mobile layer thickness ξ_τ according to [14, 18, 27],

$$\tau_\alpha(\xi_\tau) = \tau_0, \tag{6}$$

based on an empirical reference relaxation time τ_0. We have used $\tau_0 = 1$. In addition, one may introduce a third definition of thickness ξ_e according to [17, 18, 20]

$$\tau_\alpha(\xi_e) = 0.5\tau_\alpha^{\text{bulk}}, \tag{7}$$

where the mobility enhancement is gauged by the empirical prefactor 0.5 for the reduction of the relaxation time from the bulk value $\tau_\alpha^{\text{bulk}}$. Finally, from the gradient of $\tau_\alpha(z)$ close to the free surface, we define a width ξ_1 by a single-exponential form [21]

$$\tau_\alpha(z)^{-1} = (\tau_\alpha^{\text{bulk}})^{-1} + Ae^{-z/\xi_1}. \tag{8}$$

Figure 3 plots DPLM results on the thicknesses described above. As temperature decreases, we observe that ξ_τ and ξ_1 decrease while ξ_0 and ξ_e increase. The temperature dependence of ξ_τ, ξ_0, and ξ_e has been studied before. Specifically, experiments on ξ_τ [14] and MD results on ξ_τ [18], ξ_0 [31] and ξ_e [17, 18, 20] have all reported trends in good agreement with those from the DPLM in Fig. 3.

To further analyze the functional form of $\tau_\alpha(z)$, we have measured $\tau_\alpha(z)$ from supported DPLM films of thickness $h = 100$. We determine $\tau_\alpha^{\text{bulk}}$ accurately from simulations of bulk samples so that statistical errors are dominated by those of $\tau_\alpha(z)$. Figure 4 plots $\log(\tau_\alpha^{\text{bulk}}/\tau_\alpha(z))$ against z. The result strikingly resembles those from recent MD measurements [28] and can be reasonably fitted to Eq. (2) with a double-exponential form in the short range and an upward-turning tail at $z \gtrsim 30$. Nevertheless, other functional forms may also be possible (see Appendix A).

IV. WEAK SURFACE EFFECTS ON PARTICLE HOPS

Despite strong surface effects on the relaxation time, we next show that they are much weaker on particle hopping rate. At deep supercooling, particle dynamics in glass formers are dominated by activated particle hops as revealed by the emergence of secondary and higher-order peaks in the van Hove correlation function [49]. We now analyze the particle hopping rate defined by [21],

$$R_1(z) = \frac{P_{\text{hop}}(\Delta t, z)}{\Delta t}. \tag{9}$$

Here, $P_{\text{hop}}(\Delta t, z)$ denotes the probability that a particle at layer z hops during a time interval Δt and is given by

$$P_{\text{hop}}(\Delta t, z) = \langle \Theta(|\mathbf{r}_i(t+\Delta t) - \mathbf{r}_i(t)| - 1)\rangle_{t,z}, \tag{10}$$

where $\Theta(x) = 1$ if $x \geq 0$, $\mathbf{r_i}(t)$ is the position of particle i at time t and the average is taken over time t and all the particles that reside at depth z at the beginning of the time interval. We take a small time interval $\Delta t = 10^{-7}$ so that $R_1(z)$ measures the rate of all particle hops, as justified in Appendix B.

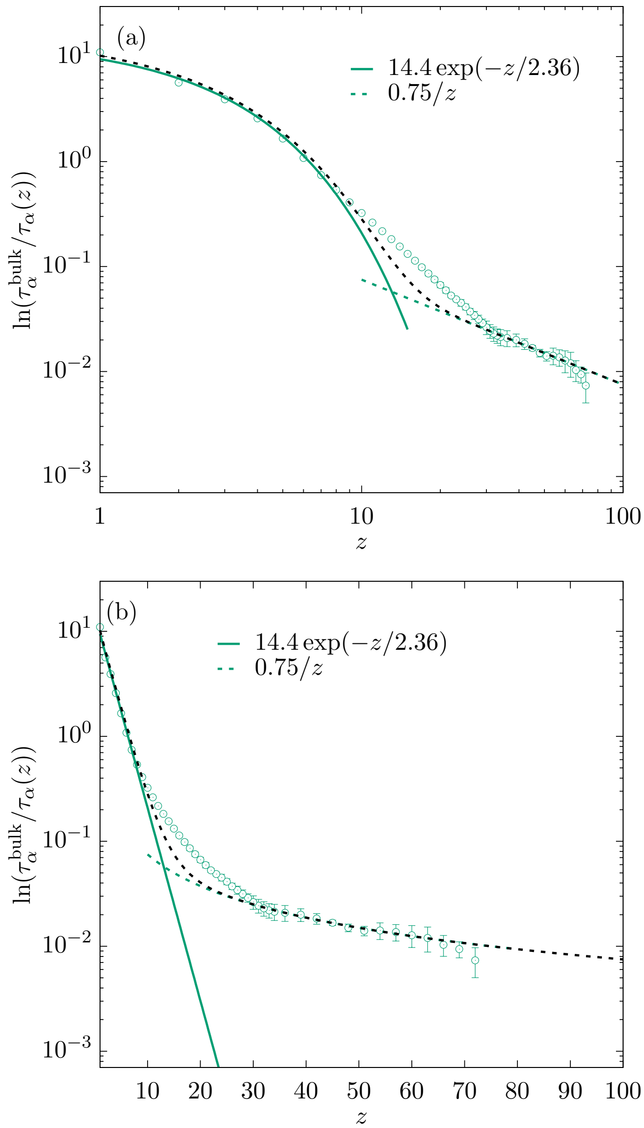

FIG. 4. Plot of $\ln(\tau_\alpha^{\text{bulk}}/\tau_\alpha(z))$ versus depth z in (a) log-log and (b) semi-log scales for a film of thickness $h = 100$ at $T = 0.21$. Two asymptotic fits to the data are also displayed. The black line indicates the sum of the two fits.

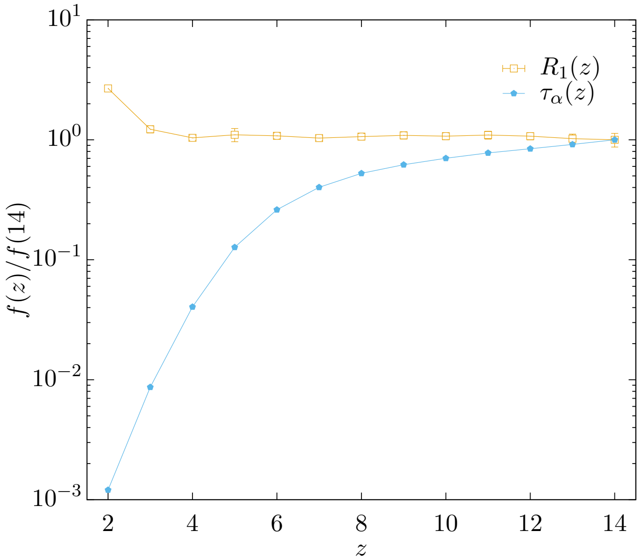

FIG. 5. Particle hopping rate $R_1(z)$ against depth z for a film of thickness $h = 15$ at temperature $T = 0.20$. Also plotted is structural relaxation time $\tau_\alpha(z)$ under the same conditions. The data is normalized by the respective quantity at $z = 14$.

Fig. 5 displays $R_1(z)$ measured against z from DPLM simulations at a low temperature of $T = 0.20$. It is compared with $\tau_\alpha(z)$ reported above and values have been normalized by their approximate bulk values taken at $z = 14$. A clear difference between the strengths of the surface effects is seen. Specifically, $\tau_\alpha(z)$ forms a gradient penetrating deep into the interior of the film up to at least $z \simeq 13$. In contrast, a gradient in $R_1(z)$ extends only up to $z \simeq 3$.

In addition, the local particle density in the DPLM also admits surface effects only very close to the surface up to $z \simeq 3$ [46], in qualitative agreement with MD results [16]. The behaviors of $R_1(z)$ and the particle density are thus similar in this regard. This can be readily understood from the short-range nature of the interactions in the DPLM so that hopping energy barriers depend only on the local particle density. The deeper penetration of surface effects on relaxation time, not accountable by the particle hopping rate, has been attributed to a breakdown of a back-and-forth hopping tendency of glass particles close to the free surface [21].

V. DISCUSSIONS

We have measured surface mobile layer thicknesses ξ_0, ξ_τ, ξ_e, and ξ_1 defined in Eqs. (2), (6), (7), and (8) respectively. Our results show that ξ_τ and ξ_1 increase with temperature, while ξ_0 and ξ_e decrease with temperature. Such opposite trends are consistent with previous studies as explained above and have been discussed based on Adam-Gibbs theory [18] and cooperative free volume theory [27]. The reproduction of these trends here further supports the applicability of the DPLM on surface-enhanced dynamics in glassy films [46, 47], in addition to several other glassy phenomena demonstrated previously [39–45].

The different properties of the various thicknesses demonstrate that a full characterization of surface enhanced mobility in general requires the whole function of layer-resolved relaxation time $\tau_\alpha(z)$, while an individual thickness reflects only certain features of the dynamics. In particular, ξ_e captures the length scale over which the system cannot maintain its bulk dynamics. It may be most closely related to the dynamic correlation length of glass in bulk, which is expected to increase as temperature decreases. In contrast, ξ_1 is a decay length of enhanced dynamics close to the surface. It dictates the length scale of the layer dominating surface flow and is thus most relevant for quantifying surface transport [10, 19]. Its value decreases as temperature decreases, signifying that transport is dominated by an increasingly thinner layer.

The ENCLE theory based on elastic interactions [35] is an insightful analytical approach used in deriving Eq. (2), providing extensive microscopic details essential for close comparisons with MD simulations and other descriptions of glass relaxation. The exponential-of-power-law tail was taken as a unique signature of a long-range elastic field [28]. The fact that it is seen in Fig. 4 for the DPLM without elasticity calls into question such a feature as decisive evidence for the elastic picture.

Both the DPLM and the ENCLE theory can account for the exponential-of-power-law tail in the relaxation time, but they essentially produce opposite predictions on the particle hopping rate. We have observed in Fig. 5 surface effects on particle hopping rate in the DPLM only in a thin surface layer in agreement with our MD results [21]. Correlations among particle hops are known to play an important role [38]. In contrast, the ENCLE theory analyzes dynamics dictated by particle activated hopping energy barriers, while correlations among hops are completely neglected. The calculations essentially extract the structural relaxation rate directly from particle hopping rate. This should predict surface effects on relaxation rate penetrating as deep as those on hopping rates. It is not clear how the ENCLE theory can be reconciled with MD results in [21].

To conclude, we have measured from DPLM simulations the temperature dependence of four definitions of surface mobile layer thickness in glassy films. Different trends are observed in agreement with previous studies and they follow from different characteristics of the layer-resolved relaxation time. We also study the relaxation time including regions deep into the film. A two-component form with a slowly decaying tail closely resembling accurate MD measurements is observed. Our results show that long-range elastic interactions, which are absent in the DPLM, are not a necessary explanation of the tail. We also demonstrate that the particle hopping rate admits much shallower surface effects in the DPLM, again in agreement with MD simulations.

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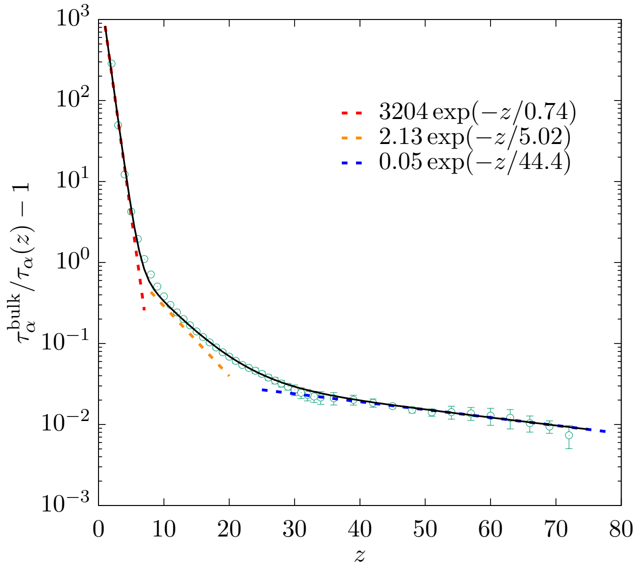

FIG. 6. $\tau_\alpha^{\rm bulk}/\tau_\alpha(z)-1$ plotted against z using the same data in Fig. 4. The red, orange and blue lines are three asymptotic exponential fits to the data. The black line, sum of the three fits, displays a good agreement with the simulation data.

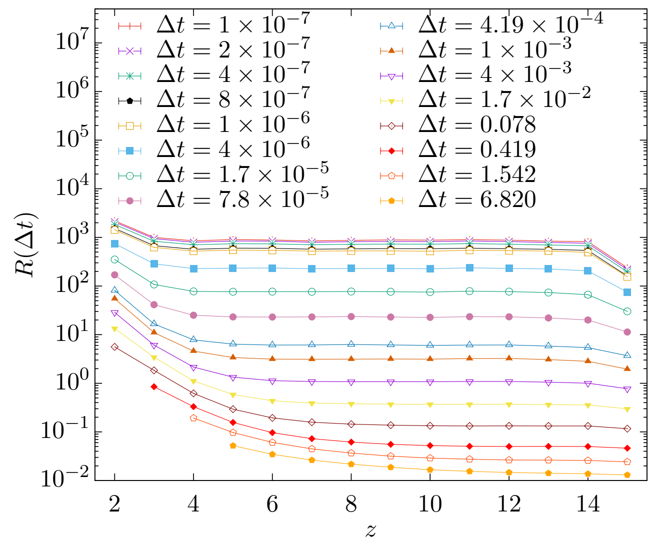

FIG. 7. Net hopping rates $R(\Delta t)$ against depth z for a film of thickness $h = 15$ at different values of Δt at $T = 0.20$.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Appendix A: Relaxation enhancement as a sum of exponentials

Figure 6 presents the data of $\tau_\alpha(z)$ from Fig. 4 fitted to an empirical sum of three exponentials

$$\tau_\alpha^{-1}(z) = (\tau_\alpha^{\rm bulk})^{-1} + A_1 e^{z/\xi_1} + A_2 e^{z/\xi_2} + A_3 e^{z/\xi_3}. \quad \text{(A1)}$$

which generalizes Eq. (8). The three exponential terms are simplified effective forms to account for possible mobility enhancement induced respectively by isolated voids exchanged with the surface, coupled voids exchanged with the surface, and all coupled voids, as motivated by simulation results using a facilitated random walk model to be reported elsewhere. At present, the finite precision of our data in Fig. 6 and the large number of fitting parameters forbid us to scrutinize Eqs. (2) and (A1) reliably.

Appendix B: Further results on particle hopping rate

A layer-resolved particle hopping rate $R_1(z)$ has been defined in Eq. (9). Following a definition applied previously to MD results [21], it is generalized to a net hopping rate

$$R(z,\Delta t) = \frac{P_{\rm hop}(\Delta t, z)}{\Delta t}, \quad \text{(B1)}$$

where $P_{\rm hop}(\Delta t, z)$ defined in Eq. (10) denotes the probability that a particle at depth z hops during time Δt. For a very small time interval $\Delta t = 10^{-7}$, the hopping rate $R_1(z)$ is restored. Figure 7 plots results on $R(z,\Delta t)$ for different Δt from DPLM simulations. Note that we have ensured that Δt is small enough by showing only data points for which $P_{\rm hop}(\Delta t, z) \leq 0.5$. The results in Fig. 7 qualitatively resemble those from MD simulations [21]. For $\Delta t \simeq 10^{-7}$, $R(z,\Delta t)$ have approximately converged and this signifies that all hops are properly counted. In contrast, $R(z,\Delta t)$ decreases with Δt at larger Δt. This is because back-and-forth hops, which are very abundant in the system, then do not contribute to $R(z,\Delta t)$, properly reflecting that they also do not contribute to the net dynamics at longer time scales as explained in [21]. In particular, the gradient in $R(z,\Delta t)$ penetrates deeper into the bulk for larger Δt. More precisely, $R(z,\Delta t)$ at $\Delta t = 6.820$ converges to its bulk value at $z \gtrsim 12$, in sharp contrast to $R_1(z)$ which converges at a much shallower depth of $z \gtrsim 3$. This exemplifies that surface effects for long-time measurements such as $R(z,\Delta t)$ for large Δt and $\tau_\alpha(z)$ penetrate deeper than short-time values such as $R_1(z)$ [21].

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